Aldehyde, Ketone and Carboxylic Acids

#Method of Preparation of Carbonyl Compounds:

1) Oxidation:→

1° R-OH —Mild OA→ Aldehyde

2° R-OH —Mild/Strong OA→ Ketone

Mild OA ⇒ PCC, PDC, Collin's Reagent, $X_2/NaOH$, NBS, $Cu, 300°C/O_2, \Delta$

Strong OA ⇒ $KMnO_4, \Delta$, $K_2Cr_2O_7, \Delta$, CrO_3/H_2SO_4

2) Ozonolysis:→

$R-CH=C(R)(R') \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} R-CHO + R-\overset{O}{\overset{\|}{C}}-R'$

3) Kucherov Reaction:→

$R-C\equiv C-H \xrightarrow[(ii)\ dil.\ H_2SO_4]{(i)\ HgSO_4} R-\overset{OH}{\overset{|}{C}}=CH_2 \rightleftharpoons R-\overset{O}{\overset{\|}{C}}-CH_3$

#Method of Preparation of Aldehyde:

1) Rosenmund Reduction:→ catalyst Poison

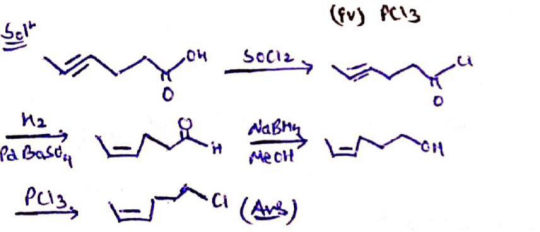
→ Lindlar's catalyst ($Pd-BaSO_4$ or $Pd-CaCO_3$) is used with H_2 on acid derivatives.

$R-\overset{O}{\overset{\|}{C}}-X \xrightarrow[BaSO_4]{H_2-Pd} R-\overset{O}{\overset{\|}{C}}-H$

[One step Reduction instead of Two step Redn]

Eg: $H_3C-C\equiv C-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OH$ —(i) $SOCl_2$, (ii) $H_2/Pd-BaSO_4$, (iii) $NaBH_4/MeOH$, (iv) PCl_3→

Sol:



2) Stephens Reduction:→

Concept: In every step of reduction, there is difference of $2e^-$. Thus, no. of reductive steps would depend on O.S. of reagent

$Sn \xrightarrow{-2e^-} Sn^{2+} \xrightarrow{-2e^-} Sn^{4+}$

Sn would reduce 2 times; Sn^{2+} would reduce only 1 time

$R-C\equiv N \xrightarrow[HCl]{+2\ SnCl_2} R-CH=NH \xrightarrow{H_2O} R-\overset{O}{\overset{\|}{C}}-H + NH_3$

Similar Redn Rxn

$R-C\equiv N \xrightarrow[(ii)\ Fe/HCl]{(i)\ Sn/HCl\ or\ Metal/Acid} R-CH_2-NH_2$ (iii) Na/EtOH

All performs two step Reduction.

Eg: CH_3-CH_3 —(i) $Cl_2/h\nu$, (ii) NaOH, (iii) $SOCl_2$, (iv) KCN, (v) $SnCl_2/HCl$, (vi) H_2O, (vii) LAH, (viii) H_2O→

Sol'n $CH_3CH_3 \xrightarrow{(i)} CH_3-CH_2-Cl \xrightarrow{(ii)} CH_3-CH_2-OH$

$\xrightarrow{(iii)} CH_3-CH_2-Cl \xrightarrow{(iv)} CH_3-CH_2-C\equiv N \xrightarrow{(v)} CH_3CH_2-\overset{HN}{\overset{\|}{C}}-H$

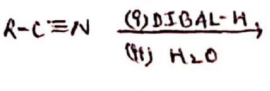
$\xrightarrow{(vi)} CH_3-CH_2-CHO \xrightarrow[(viii)]{(vii)} CH_3-CH_2-CH_2-OH$

3) Reduction by DIBAL-H:→

$R-C\equiv N \xrightarrow[(ii)\ H_2O]{(i)\ DIBAL-H} R-\overset{O}{\overset{\|}{C}}-H$

4) Gattermann Koch Synthesis:→

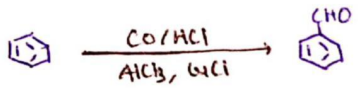
When we try to perform EAS on benzene to get Ph-CHO,



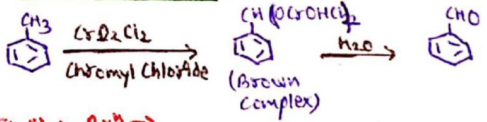
(X) → This product is not obtained

→ Because $AlCl_3$ & HCOCl reacts to form HCl & CO.

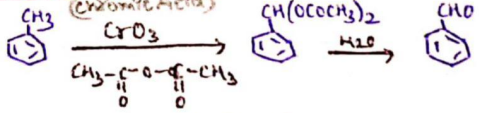
Hence, to make this rxn happen, following reactants were taken.



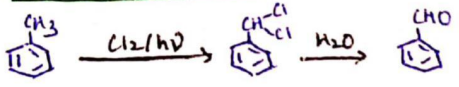
5) Etard Reaction:→



Similar Rxn ⇒



6) Side Chain Oxidation:→

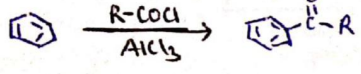


→ In JEE MAIN, attach 2-Cl atoms.

→ Although, conceptually, attaching only 1-Cl atom is correct and should be followed everywhere except during prep. of Aldehyde.

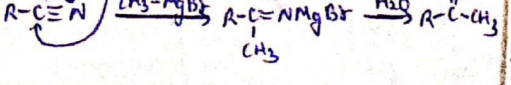
#Method of Preparation of Ketones:

1) Friedal Craft Acylation:→



2) From Nitriles:→

$R-C\equiv N \xrightarrow[(ii)\ H_2O]{(i)\ CH_3MgBr} R-\overset{O}{\overset{\|}{C}}-CH_3$



2) From Acyl Chloride:→

(i) $2R-Mg^{\delta-}X + CdCl_2 \longrightarrow R_2Cd$

(ii) $R_2Cd + 2\ R'-\overset{O}{\overset{\|}{C}}-Cl \longrightarrow 2R'-\overset{O}{\overset{\|}{C}}-R + CdCl_2$

→ Here, we directly couldn't use RMgBr because it has high reactivity & would React further.

→ While, R_2Cd is less Reactive than RMgBr

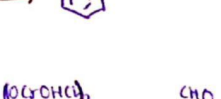
#Chemical Properties of Carbonyl Compounds:

→ The Rate of Nucleophilic attack on Carbonyl compounds depends upon

1) Electrophilicity of Carbonyl Carbon: $(-\overset{O}{\overset{\|}{C}}-Cl)$ Ke carbon pe jitna zyada δ+, Nu⁻ attack utna fast hoga.

2) Steric Hindrance: $(-\overset{O}{\overset{\|}{C}}-Cl)$ Ke carbon se jitne kam bulky group attach honge, Nu⁻ attack utna fast hoga.

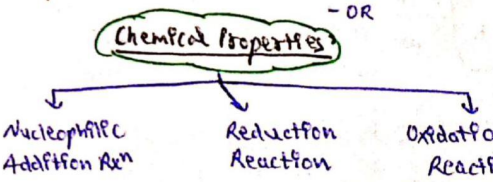
Eg: Predict order of Nucleophilic Attack:

1) 

(Due to EWG) 1 > 2 > 3 (Due to crowding)

2) $R-\overset{O}{\overset{\|}{C}}-H$ (e^- donation by +I) > $R-\overset{O}{\overset{\|}{C}}-R'$ (e^- donation by +R) > $R-\overset{O}{\overset{\|}{C}}-NH_2$ (e^- donation by +R) > $R-\overset{O}{\overset{\|}{C}}-OR'$ (e^- donation by +R)

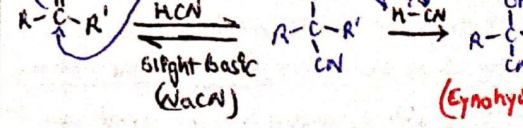
⇒ 1 > 2 > 3 > 4 $(EN)_N < (EN)_O$ thus $-NH_2$ will donate more e^- density than $-OR$



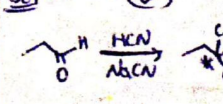
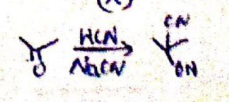
#Nucleophilic Addition Reactions:

1) Addition of HCN:→

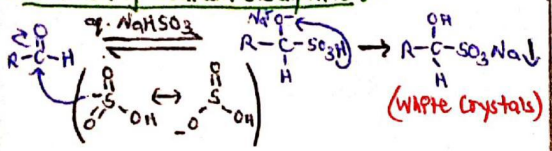
→ Base is taken with HCN, because HCN is weak acid. H^+ gets consumed by base so dissociation of HCN rises,



Eg: Find carbonyl compound of MF C_3H_6O which gives Racemic Mixture with HCN/NaCN.

Sol'n (✓)  (X) 

2) Addition of Sodium Bisulphite :→



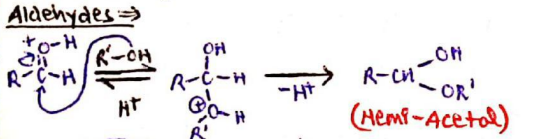
* Reversible in Nature
* Acidic medium favours the rxn
* Rxn is given only by
  - All Aldehyde (R/Ar-CHO)
  - Aliphatic methyl Ketones ($R-\overset{O}{\overset{\|}{C}}-CH_3$)
* Thus, this Rxn is used in POC to separate carbonyl compound.

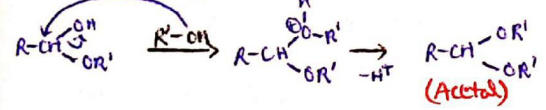
Eg: Among the following, which give white crystals with $NaHSO_3$.

1) ✓
2) ✓
3) ×
4) ✓
5) ✓
6) ✓
7) ×

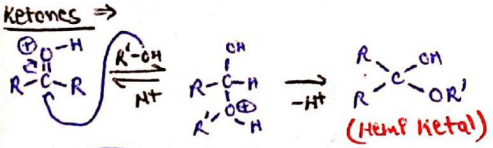
3) Addition of Alcohols :→

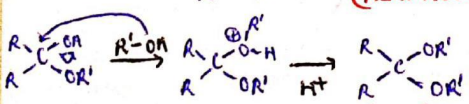
Aldehydes ⇒





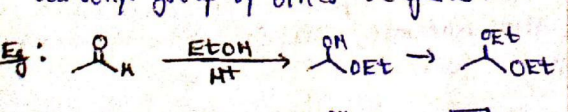
Ketones ⇒





* For formation of Acetal/Ketal, Alcohol must be present in excess amount.
* Rxn is reversible but acidic medium favours the Rxn.
* Hemi Acetal / Hemi Ketal are unstable in both acidic & basic medium.
* Acetals & Ketals are stable in basic medium
* Thus this rxn is used for protection of carbonyl group by other reagents.

Eg:

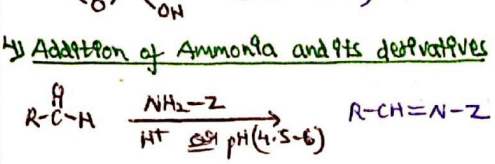


NOTE: In case of diols like ethylene glycol ($HO-CH_2-CH_2-OH$), half amount of R-OH is req. than usual to form Ketal/Acetal.

Eg: Categorise the following compounds :

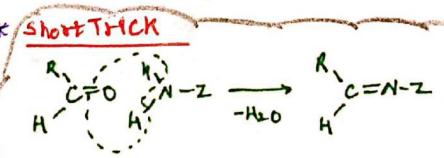
1) $CH_3-CH(OH)(OEt)$ (Hemi Acetal)
2) (Acetal)
3) (Hemi-Ketal)

4) Addition of Ammonia and its derivatives :→

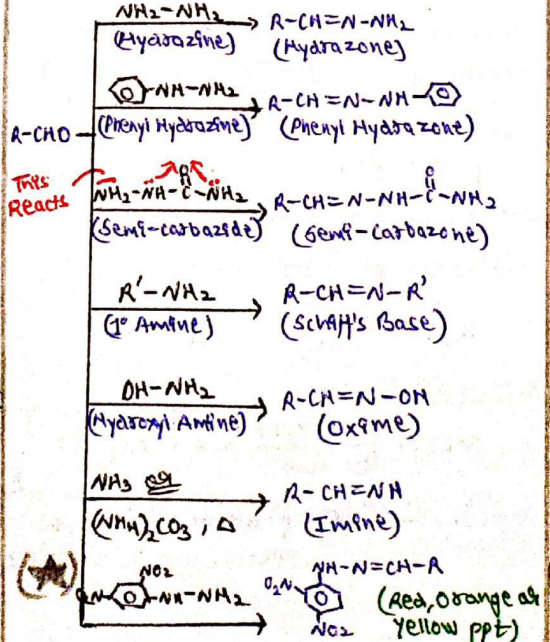


* Nucleophilic Add^n-Elimination Rxn
* Slighly acidic med. taken to ↑ electrophilicity of carbonyl carbon.

Short TRICK



Simply remove $1H_2O$ using oxygen of carbonyl & two hydrogens of NH_2-Z

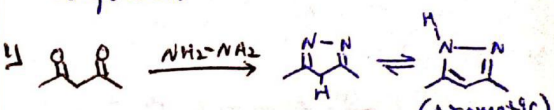


Reagent Name ⇒
* 2,4-Dinitro phenyl hydrazine
* 2,4-DNP
* Braddy's Reagent

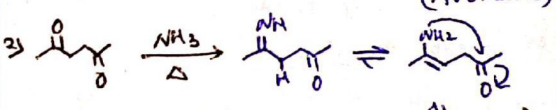
→ Product is called 2,4-Dinitro Phenyl Hydrazone. (Red Orange Yellow)

→ Product is R.O.Y coloured ppt, thus rxn used in POC to separate carbonyl compounds.

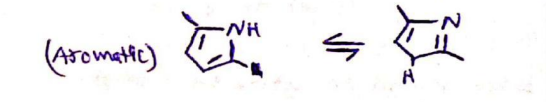
1)



2)

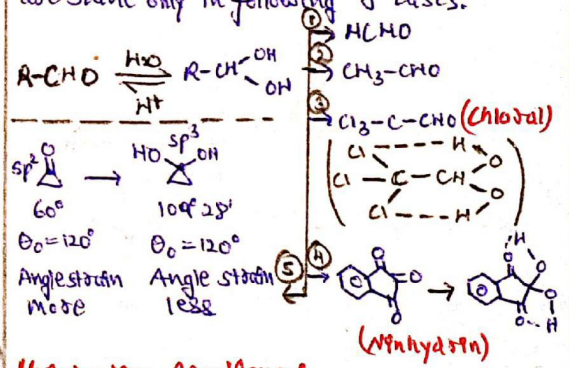


3)



5) Addition of Water (H_2O) :→

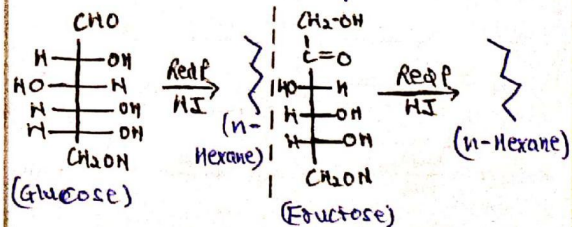
→ Due to addn of H_2O, two hydroxyl groups are attached on a single carbon and they are stable only in following 5 cases.



Reduction Reactions :

1) Red P / HI :→

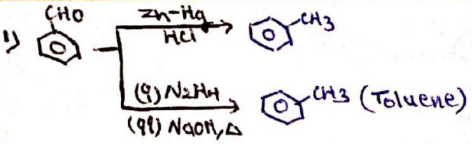
→ Very strong Reducing Agent.
→ Will reduce everything into alkane.



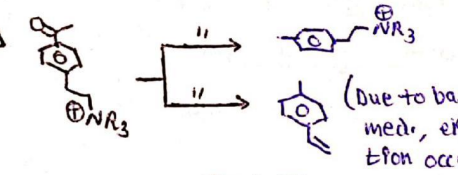
2) Clemmensen & Wolf Krshner Reduction :→

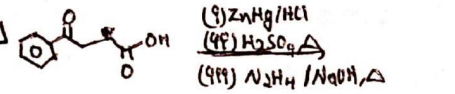
Clemmensen Red^n	Wolf Krshner Red^n
* Zn/Hg / HCl	* (i) N_2H_4 (ii) Base, Δ
* Acidic Medium	* Basic Medium
* $\diagup C=O \rightarrow \diagdown CH_2$	* $\diagup C=O \rightarrow \diagdown CH_2$
* Free Radical Mechanism	* Carbanion intermediate
* Also reacts with acid sensitive group (-OH)	* Also reacts with base sensitive groups (-Cl, -Br, etc)

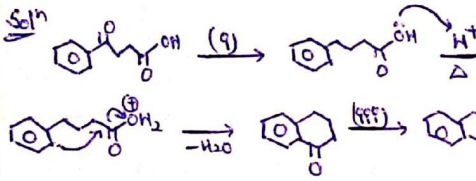
Aldehyde, Ketone & Carboxylic Acid - 2

1) 

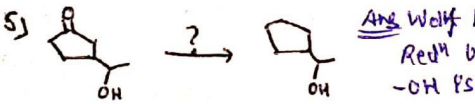
2) 

3) 

4) 

Soln 

Here H_2SO_4 is stronger acid than any organic acid, thus $Ph-CH_2CH_2CH_2-COOH$ behaved as a base and got attacked by H^+.

5) 

? ⇒ (i) N_2H_4
(ii) KOH, Δ

Ans Wolff Kishner Red^n because -OH is an acid sensitive group.

Oxidation Reaction :

1) Tollen's Test :→

Reagent → Ammonical $AgNO_3$
→ $[Ag(NH_3)_2]OH$
→ $Ag(NH_3)_2^+$
→ Ag_2O / OH^-

All means same Reagent

Tollen Formation ⇒

(i) $2AgNO_3 + 2NaOH \longrightarrow Ag_2O + 2NaNO_3 + H_2O$ (Brown)

(ii) $Ag_2O + 4NH_3 + 2NaNO_3 \longrightarrow 2[Ag(NH_3)_2]NO_3 + H_2O + 2e^-$

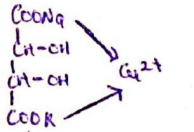
Overall Reaction ⇒

$2Ag(NH_3)_2^+ + RCHO + 3OH^- \longrightarrow 2Ag_{(s)}\downarrow + 4NH_3 + R\text{-}COO^- + 2H_2O$ (Silver Mirror)

→ carboxylic salt is obtained as product that can be made into carboxylic acid by acidification or hydrolysis.

Reacts with
① → Compound having $(-\overset{R}{\underset{}{C}}-H)$ segment (All aldehydes, $H-\overset{R}{\underset{}{C}}-OH$ acid)
② → Compounds that can form -CHO (HemiAcetal)
③ → α-Hydroxy Carbonyl 
④ → Terminal alkyne gives white ppt instead of Silver mirror with Tollen's Reagent.
($R-C\equiv C-H$, $H-C\equiv C-H$)

2) Fehling Test :→

Fehling A → $CuSO_4$ (Blue)
Fehling B → 

(Rochelle's Salt)

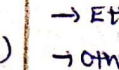
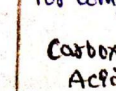
Net Reaction ⇒

$R\text{-}CHO + 2Cu^{2+} + 5OH^- \longrightarrow RCOO^- + Cu_2O + 3H_2O$ (Red ppt)

* Fehling test is given by all the groups that give positive Tollen's Test except
Aromatic Aldehydes ($Ph\text{-}CHO$)
Acetals that can form ($Ph\text{-}CHO$)
Due to lack of α-hydrogen, no Fehling Test given

NOTE : Just like Fehling's Test, there are other three Reagents.
1) Benedict's Test
2) $HgCl_2$
3) Hg_2Cl_2
Same function as Fehling.

Eg: Which of the following gives Tollen's (T) and Fehling's (F) Test:

1) $CH_3\text{-}CHO$ (T, F)
2) $Ph\text{-}CHO$ (T)
3) $HCHO$ (T, F)
4) $H\text{-}COOH$ (T, F)
5) $CH_3\text{-}CH(OH)(OCH_3)$ (T, F)
6)  (T, F)
7) $Ph\text{-}CH(OH)(OEt)$ (T)
8)  (T, F)

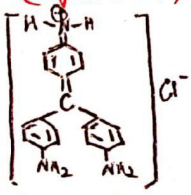
3) Schiff's Test :→

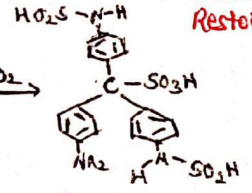
* Medium is only slightly basic, thus test is given only by free -CHO groups.

* p-Rosaniline hydrogenchloride / Fuschin is Magenta pink in colour.

* When SO_2 is passed through Fuschin, then colour disappears and upon oxidation of R-CHO, pink colour reappears.

p-Rosaniline Hydrogen Chloride (Magenta Pink) $\xrightarrow{SO_2}$ Schiff's Base (Colourless) $\xrightarrow{R\text{-}\overset{O}{\overset{\|}{C}}\text{-}H}$ $R\text{-}\overset{O}{\overset{\|}{C}}\text{-}O^-$ (Pink colour Restored)


(Fuschin)



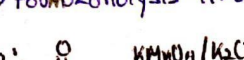
(Schiff's Base)

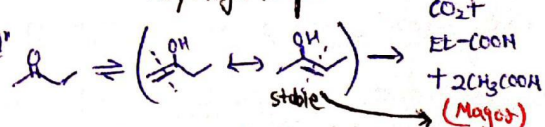
4) Popoff's Rule :→

→ Ketone can be oxidised by taking strong OA at very high temp.

→ According to this rule, oxidation is done after hypothetically forming the most stable Enol of Ketone.

→ Formozonolysis-like product of oxidation.

Eg: 

Soln 

Major product would be the product formed by most stable enol.

Physical Properties of Carbonyl Compounds :

→ Methanal (CH_2O) is gaseous at room temp.
→ Ethanal is volatile liq. at room temp
→ Other aldehydes & Ketones are solid or liquid at Room temp.

Boiling Point :→

For Comparable molecular masses,

Carboxylic Acid > Alcohol > Ketone > Aldehyde > Ether > HC

Because, for determination of BP
Molecular Mass > H-bonding > Dipole-Dipole interaction

Solubility :→

* Smaller aldehydes & Ketones (C_1-C_3) are soluble in H_2O in all proportions due to H-bonding.
* Larger aldehydes & Ketones are immiscible in water.
* Carbonyl compounds are readily soluble in organic solvents (ROR, CH_3OH, $CHCl_3$, etc)

Odour/Smell :→

* Smaller aldehydes have sharp pungent odour.
* As size rises, odour becomes fragnant.
* Many naturally occuring aldehydes & Ketones used in blending of perfumes & flavouring agents.

#Method of Preparation of Carboxylic Acids :

1) From oxidation of Alcohols & Carbonyl Compounds :→

1° R-OH $\xrightarrow[\text{Jones Reagent}]{KMnO_4 / K_2Cr_2O_7}$ R-COOH

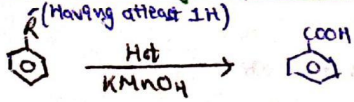
R-CHO $\xrightarrow[\text{Fehling Solution}]{\text{All above OA}}$ R-COOH

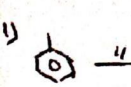
$R-\overset{O}{\overset{\|}{C}}-R' \xrightarrow[\text{Very High temp.}]{KMnO_4/K_2Cr_2O_7}$ RCOOH + - - -
(Popoff's Oxdn)

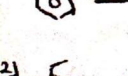
2) By Oxidative Ozonolysis :→

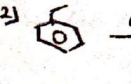
$R-CH=CH-R' \xrightarrow[H_2O_2/H_2O]{O_3}$ R-COOH + R'-COOH

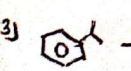
3) From Oxidation of Alkylated Benzene :→

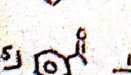


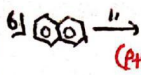
1) 

2) 

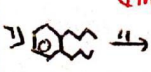
3) 

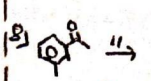
4) 

5) 

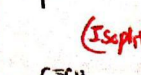
6) 

(Phthalic Acid)

7) 

8) 

(Isophthalic Acid)

9) 

(Terphthalic Acid)

4) From Hydrolysis :→

$R-C\equiv N \xrightarrow[\text{Partial Hydro.}]{H^+/H_2O} R-\overset{O}{\overset{\|}{C}}-OH$ (95% H_2SO_4; $\xrightarrow{OH^-/H_2O}$)

To make acid, we take 95% H_2SO_4 for complete Hydrolysis

$R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[OH^-/H_2O]{H^+/H_2O} R-\overset{O}{\overset{\|}{C}}-OH$

$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[OH^-/H_2O]{H^+/H_2O} R-\overset{O}{\overset{\|}{C}}-OH$

$R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R \xrightarrow[OH^-/H_2O]{H^+/H_2O} 2R-\overset{O}{\overset{\|}{C}}-OH$

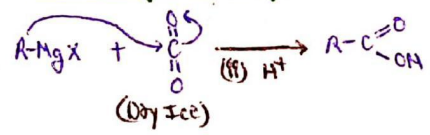
Esters ⇒

$R-\overset{O}{\overset{\|}{C}}-OR' \xrightarrow{H^+/H_2O} R-COOH + R'-OH$

$R-\overset{O}{\overset{\|}{C}}-OR' \xrightarrow{OH^-/H_2O} R-COO^- + R'OH$

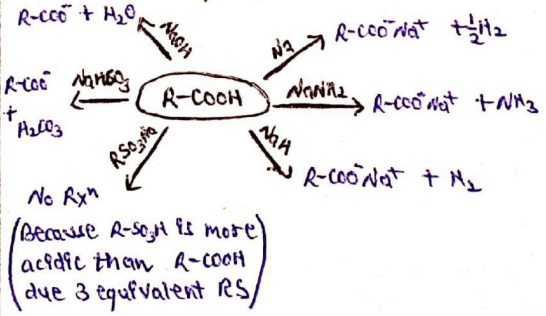
↳ **Saponification :** Hydrolysis of ester in basic medium is called saponification

5) From Grignard Reagent :→



#Chemical Properties of Carboxylic Acid :

1) Acidic strength of Acid :→ ($pK_a = 4.2$)



(Because R-SO_3H is more acidic than R-COOH due 3 equivalent EWG)

2) Rxn with PCl_3, PCl_5, $SOCl_2$:→

$R-COOH \xrightarrow{PCl_3/PCl_5/SOCl_2} R-\overset{O}{\overset{\|}{C}}-Cl$

3) Anhydride Formation :→

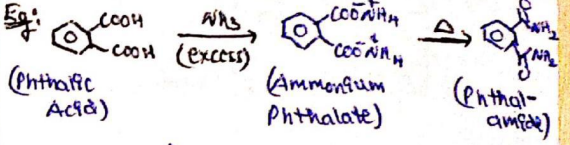
$R-\overset{O}{\overset{\|}{C}}-OH + HO-\overset{O}{\overset{\|}{C}}-R \xrightarrow[\text{or } P_2O_5, \Delta \text{ or Dehydrating Agent}]{H^+, \Delta} R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$ (− H_2O)

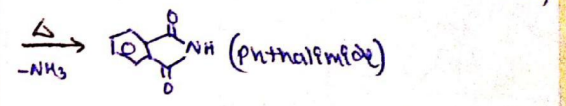
4) Esterification :→

$R-COOH \xrightarrow[\Delta]{R'OH} R-\overset{O}{\overset{\|}{C}}-OR'$

5) Reaction with Ammonia :→

$R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[(Base)]{NH_3} R-\overset{O}{\overset{\|}{C}}-O^-NH_4^+ \xrightarrow[-H_2O]{\Delta} R-\overset{O}{\overset{\|}{C}}-NH_2$

Eg: 



6) Decarboxylation Reactions :→

(i) Sodalime Decarboxylation ⇒

$R-COONa \xrightarrow[\Delta]{NaOH, CaO \ (1:3)} R-H$

(ii) Kolbe Electrolysis ⇒

$R-COONa \xrightarrow{\text{Electrolysis}} R-R$

7) Hell Vohlard Zelensky (HVZ) Reaction :→

* Here, Red P is taken in very small, catalytic amount while X_2 is taken in excess.
* Therefore, very less PX_3 is formed and the product is NOT $R-CH_2-COX$.

$R-CH_2-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[X_2 \ (Cl_2, Br_2)]{\text{Red P}} R-\underset{X}{CH}-\overset{O}{\overset{\|}{C}}-OH$

(α-position pe X lagana hai)
⇒ α-H must be present for Rxn to take place

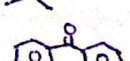
8) Dry Distillation of Calcium Salt of Acid :→

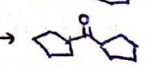
a) $A-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[\text{or CaO}]{Ca(OH)_2} \left(R-\overset{O}{\overset{\|}{C}}-O\right)_2Ca \xrightarrow[\text{Distillation}]{\text{Dry}} R-\overset{O}{\overset{\|}{C}}-R + CaCO_3$

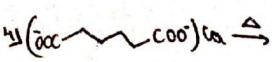
b) $\left(R'-\overset{O}{\overset{\|}{C}}-O\right)_2Ca + \left(R-\overset{O}{\overset{\|}{C}}-O\right)_2Ca \xrightarrow[\text{Distillation}]{\text{Dry}} R-\overset{O}{\overset{\|}{C}}-R' + CaCO_3$

→ Same reaction can be performed with MnO/Δ as well.

1) $(CH_3COO)_2Ca \xrightarrow{\Delta}$ 

2) $(Ph-COO)_2Ca \xrightarrow{\Delta}$ 

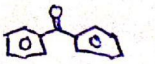
3) $\left(\text{  }-COO\right)_2Ca \xrightarrow{\Delta}$ 

4)  $\xrightarrow{\Delta}$ 

5) $(Ph_2COO)_2Ca + (CH_3COO)_2Ca \xrightarrow{\Delta}$ 

6) $CH_3COOH \xrightarrow[\Delta]{MnO}$ 

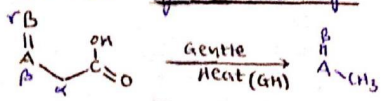
7) $Ph-COOH \xrightarrow[\Delta]{MnO}$ 

8) $Ph-COOH + CH_3COOH \xrightarrow[\Delta]{MnO}$ 

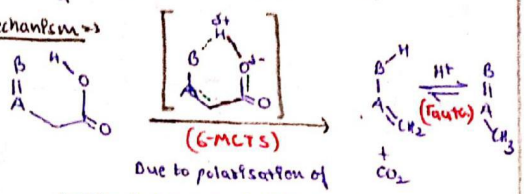
Heating Effects :

(I) β-γ Unsaturated Carboxylic acid :→

→ Such acids decarboxylate (release CO_2) even on 'gentle heating'.

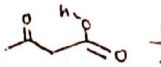


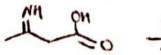
Mechanism →

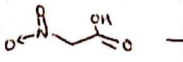


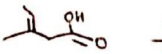
Due to polarisation of O-H bond, the π-bond A=B also gets polarised, resulting in 6-MCTS

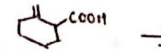
* The -H of acid is attached to γ-atom
* The product $\left(\begin{matrix}A=B\\ CH_3\end{matrix}\right)$ will be obtained only if $(H-B-A=CH_2)$ can show 'tautomerism'

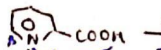
1) 

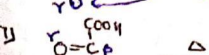
2) 

3) 

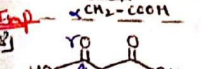
4) 
(No tautomerism)

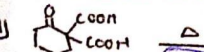
5) 
(No tautomerism)

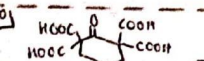
6) 
(Yaha, lp Ko γ-species Ki toh toeat Karenge)

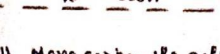
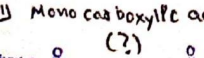
7) 
(Pyruvic acid)

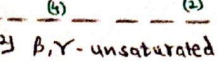
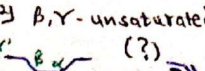
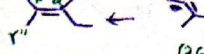
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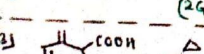
8) 
(Geminal dicarboxylic acid) — Smaller mono-carboxylic acid is obtained

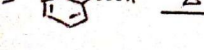
9) 
(Total 2 mol CO_2 released)

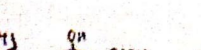
10) 

11) Mono carboxylic acid 
(?) 
Total 8 such acids are possible

12) β,γ-unsaturated acid 
(?) 
(2 G.I) or (1) 

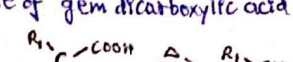
13) 
(Toluene)


(2 G.I + 2 O.I)
Total 7 such acids

14) 
Tautomerism

(II) Heating of Dicarboxylic acid :→

→ Removal of -COOH (decarboxylation) takes place in case of gem dicarboxylic acid & CO_2 is released

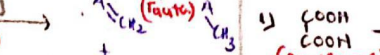


→ In case of ~~geminal~~ other dicarboxylic acids

Oxalic acid ⇒ Formic acid (HCOOH)

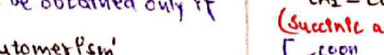
Where 5 or 6 member ring possible ⇒ Cyclic anhydride

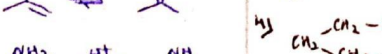
Where 5 or 6 member ring not possible ⇒ Cyclic Ketone

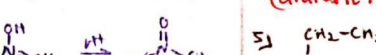
1) 
(Oxalic acid)

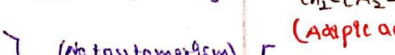
2) 
(Malonic acid)

3) 
(Succinic acid)

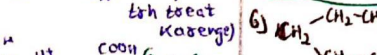


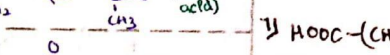
4) 
(Glutaric Acid)

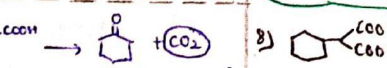
5) 
(Adipic acid)

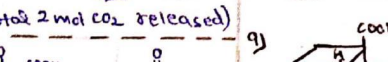


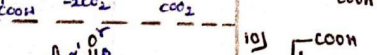
Yaha 7 membered cyclic anhydride ki possibility thi, isly instead of anhydride, yaha Ketone ban gya

6) 

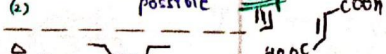
7) 

8) 

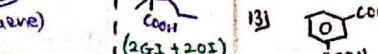
9) 

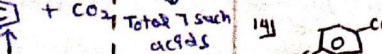
10) 
(Maleic acid) (Cyclic anhydride) (Maleic Anhydride)

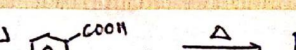
Imp

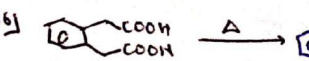
11) 
(Fumaric acid) — No cyclic anhydride formation due to trans geometry.

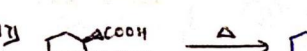
12) 
(Phthalic acid) (Phthalic anhydride)

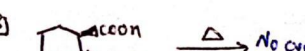
13) 

14) 

15) 

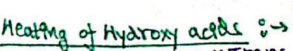
16) 

17) 

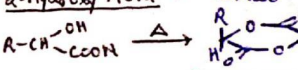
18) 
No cyclic anhydride formation

(III) Heating of Hydroxy acids :→

V. Imp

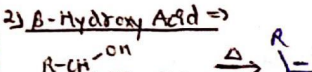
1) α-Hydroxy Acid ⇒ 
(Cos present)

Mechanism ⇒



* H^+ not attack on -OH of COOH bcz lp in Resonance
* H^+ not attack on -OH bcz dehydration of alcohol will result in 5t near $\left(-\overset{H}{\underset{5t}{C}}-\right)$ which is unstable

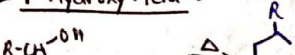
* After attack of H^+ on $\left(\overset{H}{C}\right)$, the internal nucleophile alcohol (-OH) does not attack bcz 3 member ring was getting formed, therefore dimerization occurs.

2) β-Hydroxy Acid ⇒ 

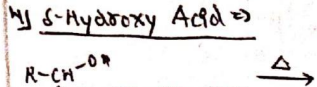
Here, simple dehydration of alcohol occured bcz

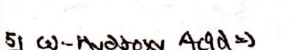
(i) Due to dehydration, 5t is not formed on adjacent carbon of $\left(\overset{5t}{C}=O\right)$

(ii) Double bond formed is in conjugation

3) γ-Hydroxy Acid ⇒ 
(Lactone)

Here, attack by internal nu- (-OH) will take place due to 5 member ring.

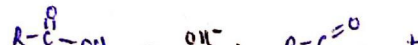
4) δ-Hydroxy Acid ⇒ 
(Lactone) cyclic ester

5) ω-Hydroxy Acid ⇒ 

ω-hydroxy acids or greater acids form poly-ester polymers by condensation Reaction.

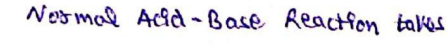
→ Majority of the Name Reactions can be understood by the following cases.

Case-01: Acid is Present ⇒



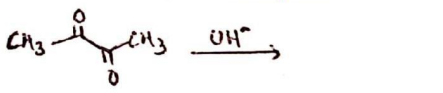
Normal Acid-Base Reaction takes place.

Case-02: Acid derivative ⇒

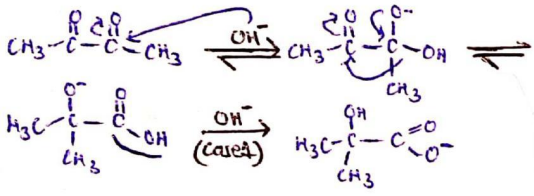


* lg ⇒ F, Cl, Br, -OR, -$\overset{O}{\overset{\|}{C}}$-OR, $-NR_2$, $-CX_3$, etc.
* 2 mole of OH^- consumed in Rxn
* Last step (neutralisation) is driving force.

Case-03: α,β-dicarbonyl ⇒

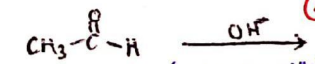


Mechanism ⇒

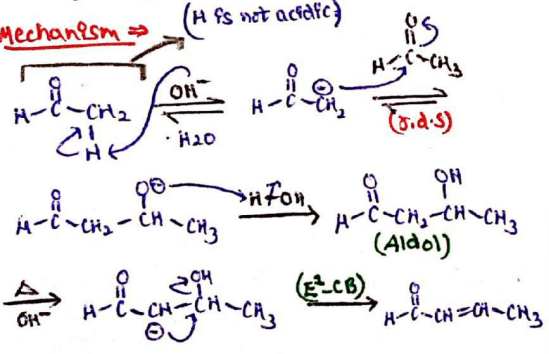


* carbonyls should only be Ketone or aldehyde
* Last step (Neutralisation) is driving force.

Case-04: When α-H present at Carbonyl ⇒ (Aldol Condensation Rxn)



Mechanism ⇒



* Reactions Involved:
  1) Carbanion Formation
  2) Nucleophilic Addition
  3) Acid-Base Rxn
  4) Carbanion Formation
  5) Elimination Dehydration (E^1_{cB})

* Nucleophilic Addition is r.d.s
  - Rate of aldol Formation: $R-\overset{O}{\overset{\|}{C}}-H > R-\overset{O}{\overset{\|}{C}}-R$
  - In case of Cross Aldol, Nucleophilic attack will take place on less crowded carbonyl.

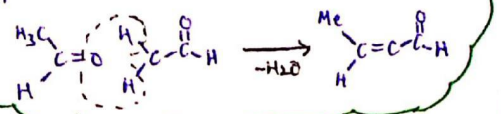
Carbonyl Name Reactions

* Base catalysts that can be taken ⇒

NaOH / KOH / $NaHCO_3$ / Na_2CO_3 / $Ba(OH)_2$ / $Ca(OH)_2$

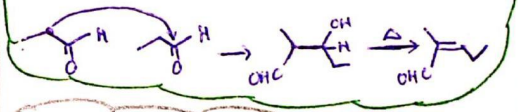
Shortcut-1

Simply Remove water using 'O' of carbonyl and α-H. It directly gives condensation product.



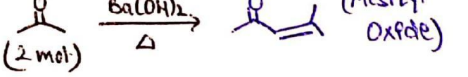
Shortcut-2

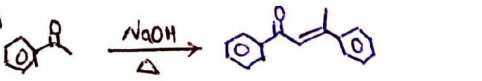
* Identify the α-position, from which carbanion will be formed.
* In case of cross aldol, choose α-position of bulkier carbonyl for carbanion
* Then simply attach it to carbonyl carbon of other molecule and proceed.

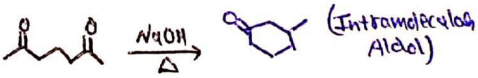


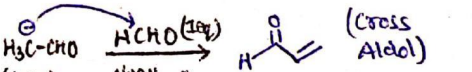
NOTE: In first step of aldol Rxn, the -H got removed because it was the last possibility for Rxn to move forward.

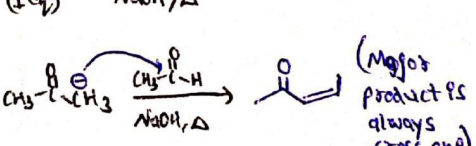
It does not get removed because it was acidic.

1) 

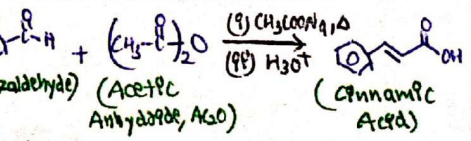
2) 

3) 

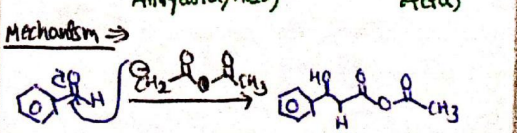
4) 

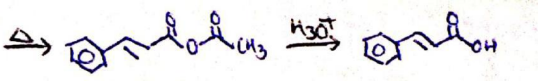
5) 

#Perkin Condensation Reaction:



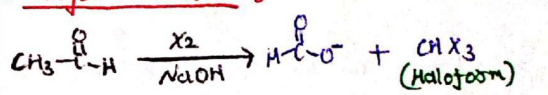
Mechanism ⇒



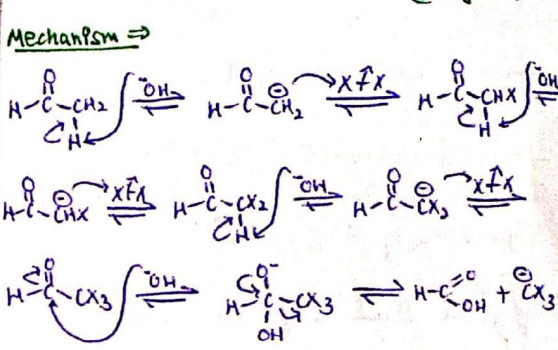


* Mechanism is similar like aldol Rxn.

#Haloform Reaction:



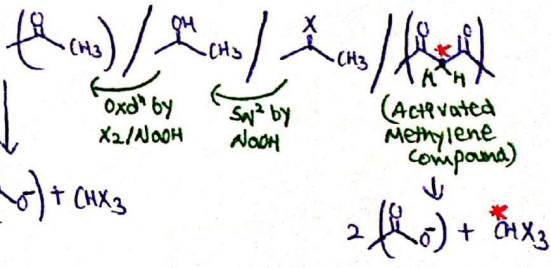
Mechanism ⇒



* Once the Rxn gets started, it stops only after removal of $CX_3^\ominus$ leaving group because with addn of each -X atom, acidity of α-H keeps increasing.
* Thus, removal of α-H for carbanion formation keeps becoming more favourable and Rxn proceeds forward.

* Reagents ⇒
  - NaOH / X_2
  - NaOX (Sodium Hypohalite)
  - $OX^\ominus$
  - Bleaching Powder ($\overset{\ominus}{O}Cl$)

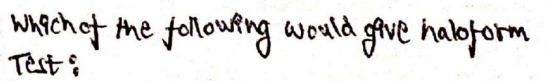
* Reaction is shown by compounds having any of following groups:



Oxd^n by X_2/NaOH

S_N2 by NaOH

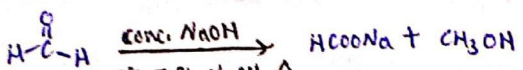
(Activated Methylene compound)



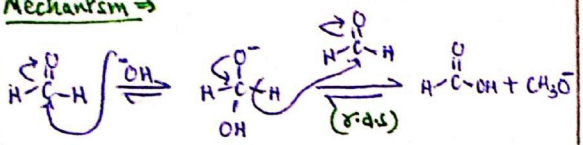
Eg: Which of the following would give haloform Test:

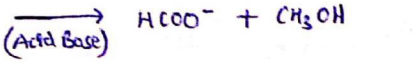
1) CH_3-CHO ✓
2) H-CHO ✗
3) Acetone ✓
4) Ethanol ✓
5)  ✓
6)  ✓
7)  ✗
8)  ✓
9)  ✓
10)  ✓
11)  ✓
12) Cl_3C-CHO ✓

Cannizaro Reaction :



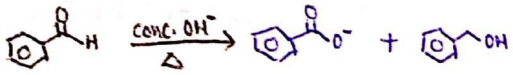
Mechanism ⇒

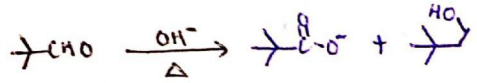




→ Disproportionation Reaction

→ Occurs only in absence of α-H and presence of strong Alkali

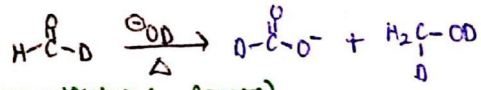
1) 

3) 

(Cross Cannizzaro)

3) $HCHO + PhCHO \xrightarrow[\Delta]{OH^-} HCOO^- +$ 

Jaha attack karna aasan ho (less crowded), woh oxidise hoga aur jaha crowding zyada ho, woh reduce hoga,

4) 

(Intramolecular Cannizzaro)

5) 